The Cosmological Constant

Directed Reading Program in Mathematics

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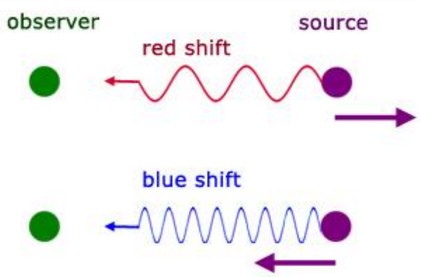
Throughout this presentation we assume a homogeneous and isotropic universe. This essentially means that there are no preferred locations or directions in space so on a large scale no matter where you look or in which direction you are looking, the universe will appear and act the same.

Hubble's Constant

Astrologist Edwin Hubble is known for finding the distances and redshift/blueshift for 20 galaxies and comparing them.



$$z = \frac{\lambda_{\text{ob}} - \lambda_{\text{em}}}{\lambda_{\text{em}}}$$

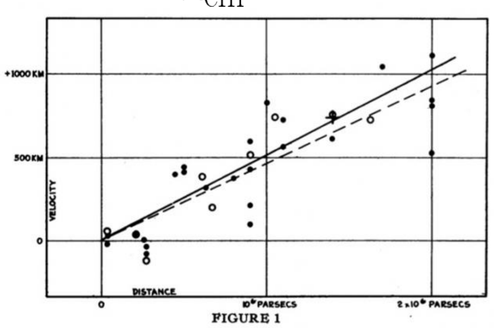


FIGURE 1
Velocity-Distance Relation among Extra-Galactic Nebulae.

After graphing velocity versus position, his data led him to discover the linear relationship between distance and redshift which we call Hubble's Law:

$$z = \frac{H_0}{c} r$$

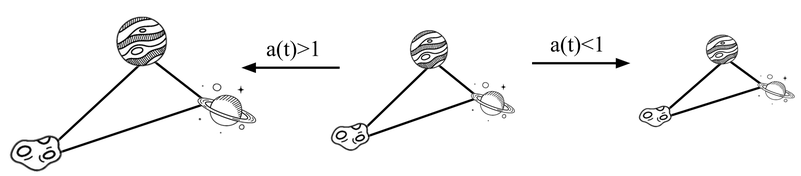
The Universal Scale Factor

After Hubble's discovery, physicists had to come up with a way to explain an expanding universe. They did so by introducing a scale factor a(t) that shrinks or expands distances over time. Since we assume homogeneousness and isostropicnous, this a(t) can be applied to any and all objects in the universe.

$$\vec{r}(t) = a(t)\vec{r}_0$$

$$\frac{d}{dt} \qquad \frac{d^2}{dt^2}$$

$$\vec{v}(t) = \dot{a}(t)\vec{r}_0 = \frac{\dot{a}(t)}{a(t)} r(t) \qquad \vec{a}(t) = \ddot{a}(t)\vec{r}_0 = \frac{\ddot{a}(t)}{a(t)} r(t)$$



Derivation of the Acceleration Equation

Take Friedmann's equation and multiply it by a²:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3c^2}\varepsilon(t) - \frac{\kappa c^2}{R_0^2}\frac{1}{a(t)^2}$$

Take the time derivative:

$$\dot{a}^2 = \frac{8\pi G}{3c^2}\varepsilon a^2 - \frac{\kappa c^2}{R_0^2}$$

Divide by 2aȧ:

$$2\dot{a}\ddot{a} = \frac{8\pi G}{3c^2}(\dot{\varepsilon}a^2 + \varepsilon 2a\dot{a})$$

From the fluid equation we can substitute the following relation:

$$\dot{\varepsilon}\frac{a}{\dot{a}} = -3(\varepsilon + P)$$

$$\frac{\ddot{a}}{a} = \frac{4\pi G}{3c^2}(\dot{\varepsilon}\frac{a}{\dot{a}} + 2\varepsilon)$$

The result is the Acceleration Equation which tells us how the speed of the universe's expansion/contraction changes with time. In this form, ä/a will always be negative

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2}(\varepsilon + 3P)$$

Why was the Cosmological Constant Originally Created?

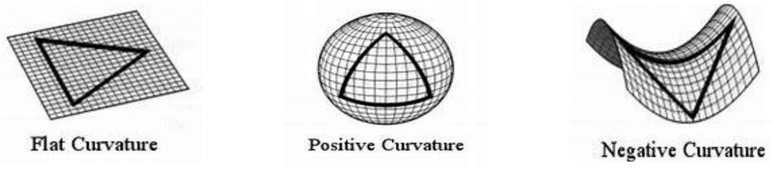
Before Hubble's discoveries, Albert Einstein had no reason to believe that we lived in anything but a static universe. However, his theory predicted this was impossible. Therefore he added a term to his equations, the Cosmological Constant. If tuned properly a static universe was now possible.

*Einstein famously called lambda the "biggest blunder" of his career

Shape of the Universe

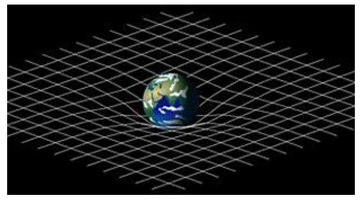
The intrinsic curvature of the universe affects the evolution!

$\kappa = 0$ $\qquad$ $\kappa = 1$ $\qquad$ $\kappa = -1$



Einstein's Field Equations

The following equation is Einstein's Field equation and it gives the mathematical relationship between spacetime curvature and energy density.



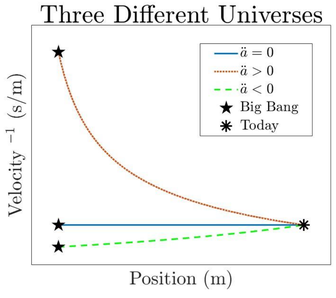
$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

After a bit of manipulation and rewriting we can go from Einstein's field equation to the Friedmann Equation.

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3c^2}\varepsilon(t) - \frac{\kappa c^2}{R_0^2}\frac{1}{a(t)^2}$$

What is the Purpose of the Cosmological Constant now?

- "Hubble time" predicts a universe younger than the Earth
- To make the universe older, we need some magical constant to change acceleration and make ä/a positive



$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2}(\varepsilon + 3P) + \frac{\Lambda}{3}$$

Lambda can be interpreted as a description of dark energy but many of its physical properties remain a mystery.